

MULTI-LABEL CHEST X-RAY CLASSIFICATION

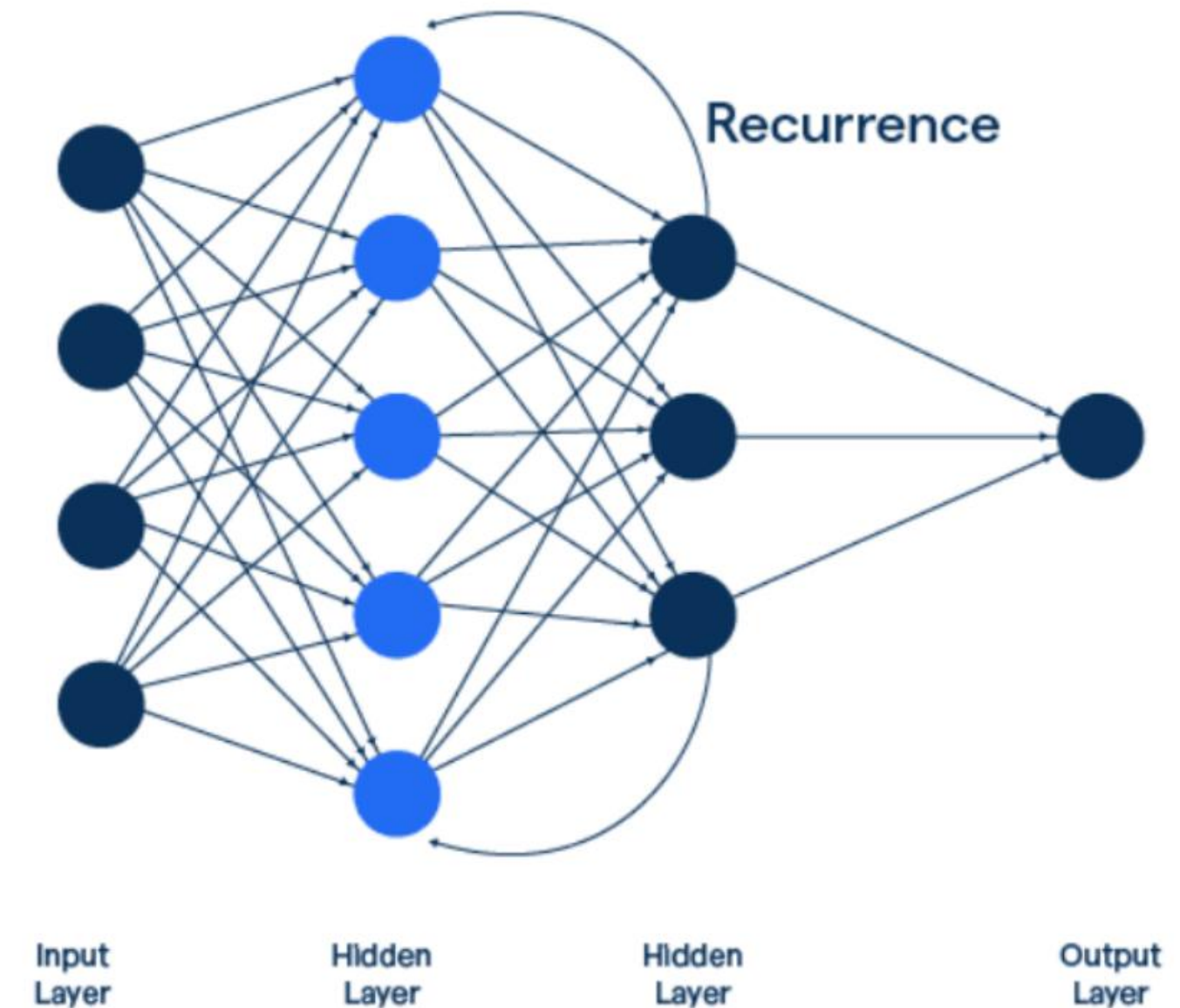
USING CONVNEXT-TINY VARIANTS FOR THORACIC DISEASE DETECTION

NEURAL NETWORKS FINAL PROJECT

Biomedical Imaging | Yachay Tech University

PROBLEM AND MOTIVATION

- ✓ **Clinical Demand:** Chest X-rays are primary screening tools for thoracic diseases [1].
- ✓ **Complexity:** Manual interpretation is time-intensive and subject to inter-observer variability.
- ✓ **Multi-label Nature:** A single patient often presents multiple co-occurring findings.
- ✓ **Objective:** Develop an AI decision-support system to streamline preliminary screening.



GENERAL AND SPECIFIC OBJECTIVES

GENERAL OBJECTIVE

Develop and evaluate a neural-network-based system for multi-label classification of thoracic findings in frontal chest X-rays using modern ConvNet architectures.

KEY SPECIFIC OBJECTIVES

- ✓ Partition CheXpert dataset using patient-wise splitting [1].
- ✓ Optimize ConvNeXt-Tiny variants for medical feature extraction [2].
- ✓ Implement multi-label probability averaging ensemble.

KEY SPECIFIC OBJECTIVES (CONT.)

- ✓ Apply Grad-CAM for clinical interpretability [4].
- ✓ Evaluate performance via macro-averaging and optimized thresholds.

STATE OF THE ART

Author / Year	Dataset	Architecture	Contribution
Irvin et al. (2019) [1]	CheXpert	CNN Baseline	Established CheXpert dataset and uncertainty labeling.
Liu et al. (2022) [2]	ImageNet	ConvNeXt	Modernized CNN blocks to compete with Transformers.
Lin et al. (2017) [3]	Benchmarks	Focal Loss	Addressed class imbalance by down-weighting easy examples.
Selvaraju et al. (2017) [4]	Visual Tasks	Grad-CAM	Visual interpretability via gradient-based localization.
Pham et al. (2020) [5]	CheXpert	CNN Ensemble	Utilized label smoothing and hierarchical dependencies.

Our project integrates these advancements into a reproducible ConvNeXt-Tiny pipeline focused on patient-wise reliability.

DATABASE: KAGGLE CHEXPRT-V1.0-SMALL

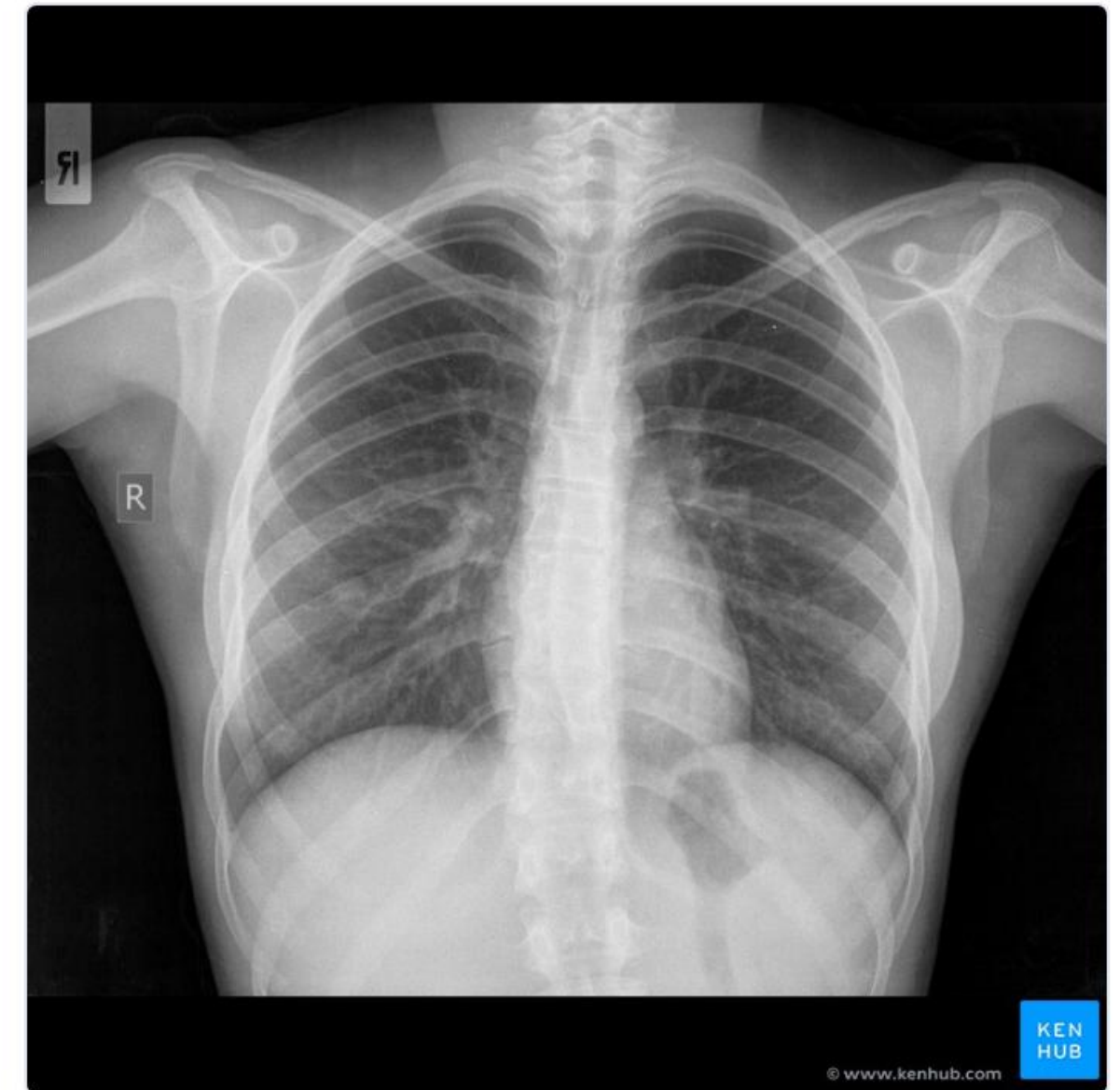
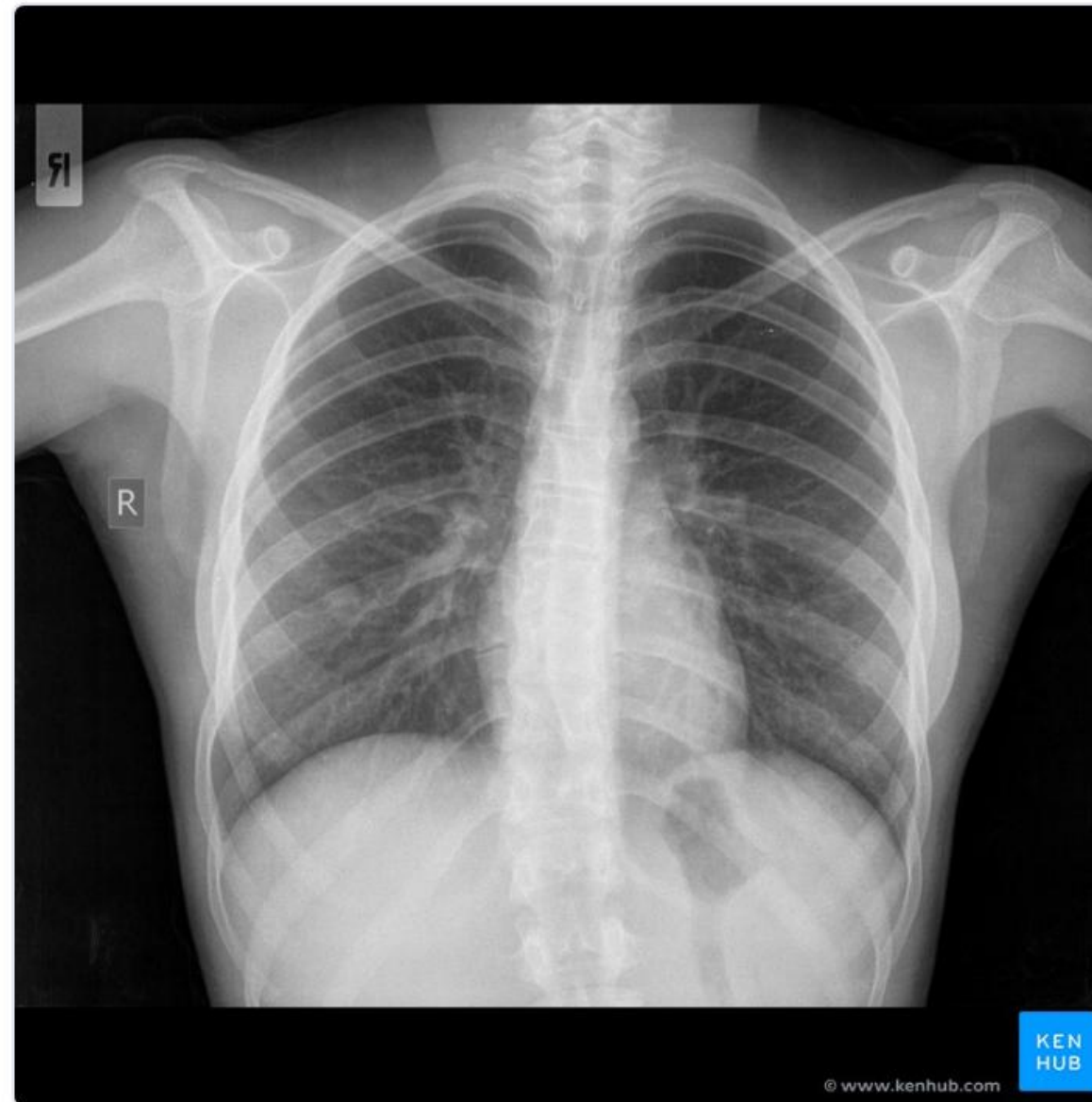
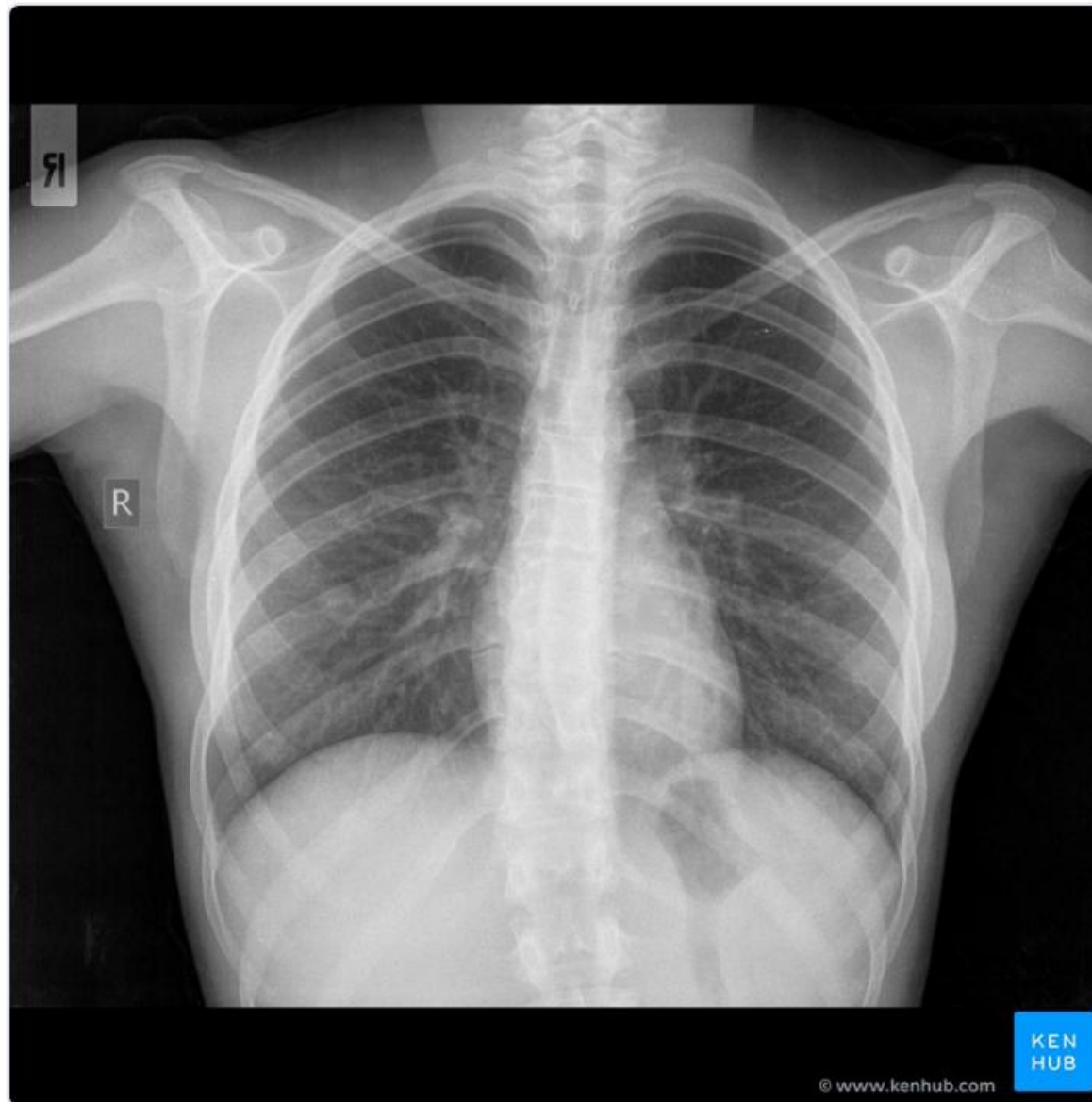
TARGET FINDINGS

- ✓ Atelectasis
- ✓ Cardiomegaly
- ✓ Consolidation
- ✓ Edema
- ✓ Pleural Effusion

Train Set	Validation Set	Test Set
80,008 Images 27,222 Patients	10,017 Images 3,500 Patients	10,000 Images 3,564 Patients

CRITICAL DESIGN: Patient-wise splitting ensures 0 patient overlap across subsets.

FRONTAL DATASET SAMPLES



Input Dimensions: 320 x 320 | Normalization: ImageNet Stats

SYSTEM PIPELINE



PREPROCESSING

Consistent normalization and aspect-ratio preservation for stable convergence.



INTERPRETABILITY

Parallel Grad-CAM path for visual verification of high-probability findings.

ENSEMBLE ARCHITECTURE



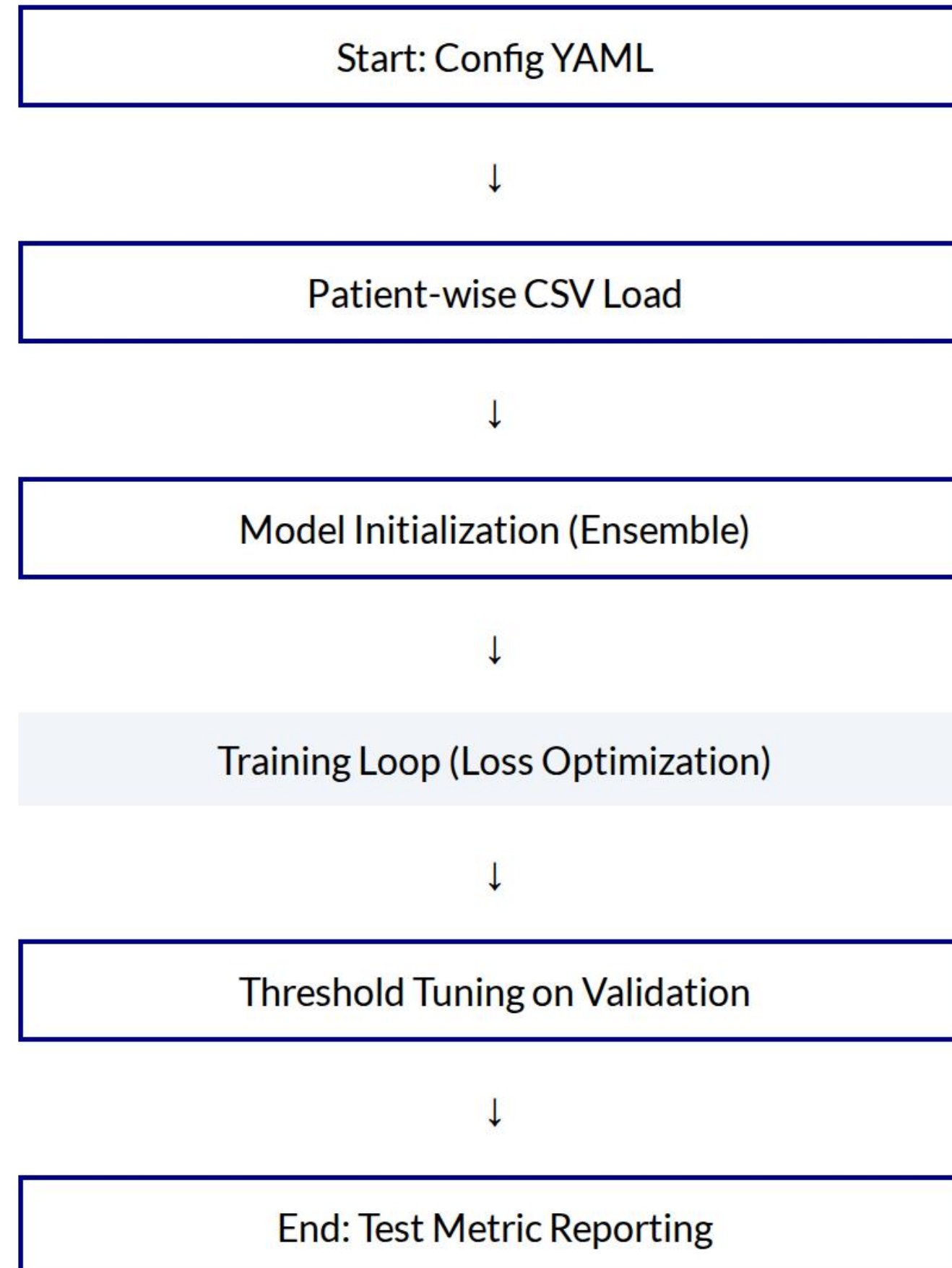
All branches utilize the **ConvNeXt-Tiny** backbone modernized with depthwise convolutions and GeLU activation.

TRAINING CONFIGURATION

Batch Size	8
Image Size	320 x 320
Optimizer	AdamW
Initial LR	1e-4
Scheduler	ReduceLROnPlateau

Frozen Head Stage	1-3 Epochs
Fine-tuning Stage	Up to 35 Epochs
Early Stopping	Patience 6
Class Weights	Calculated per Positive Label
Augmentation	Random Horizontal Flip

SYSTEM FLOWCHART



EVALUATION METRICS

AUC-ROC

Discrimination capacity
independent of thresholds.

MACRO F1

Harmonic mean of precision and
recall for all classes.

RECALL

Ability to detect all positive cases
(High priority in medicine).

DECISION FORMULA (SIGMOID ENSEMBLE)

$$\hat{P} = \frac{1}{N} \sum_{i=1}^N \sigma_i (x)$$

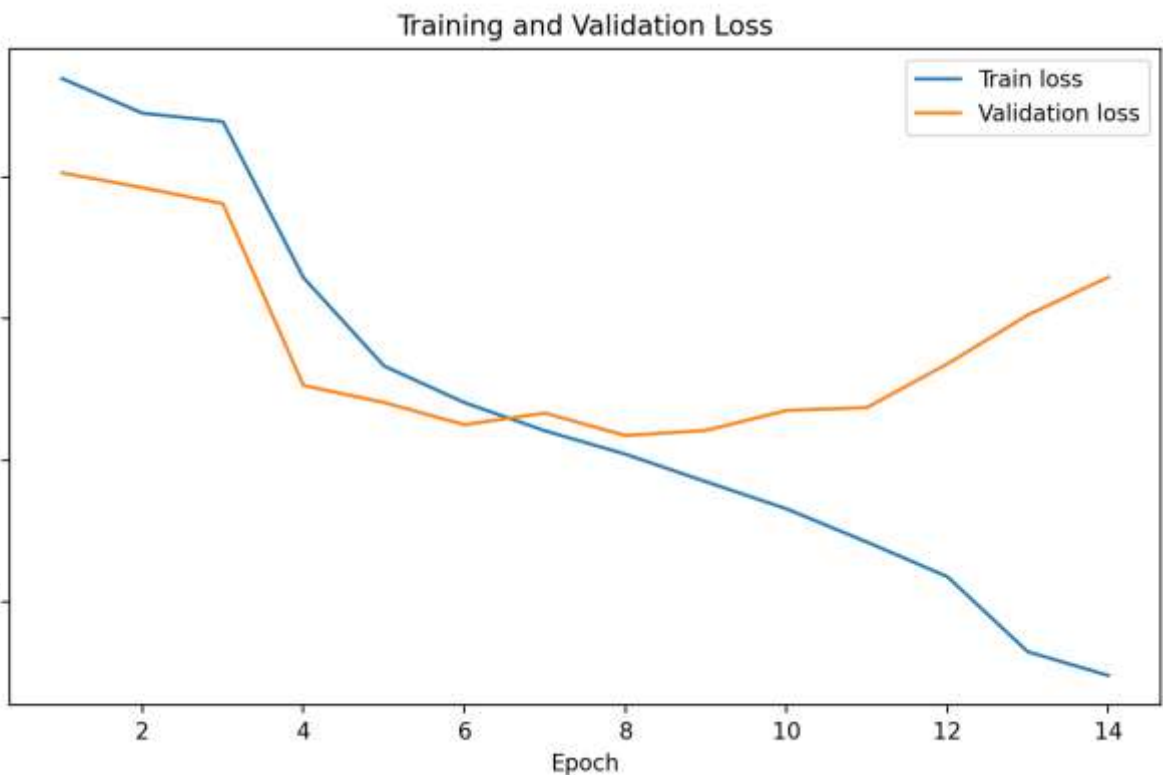
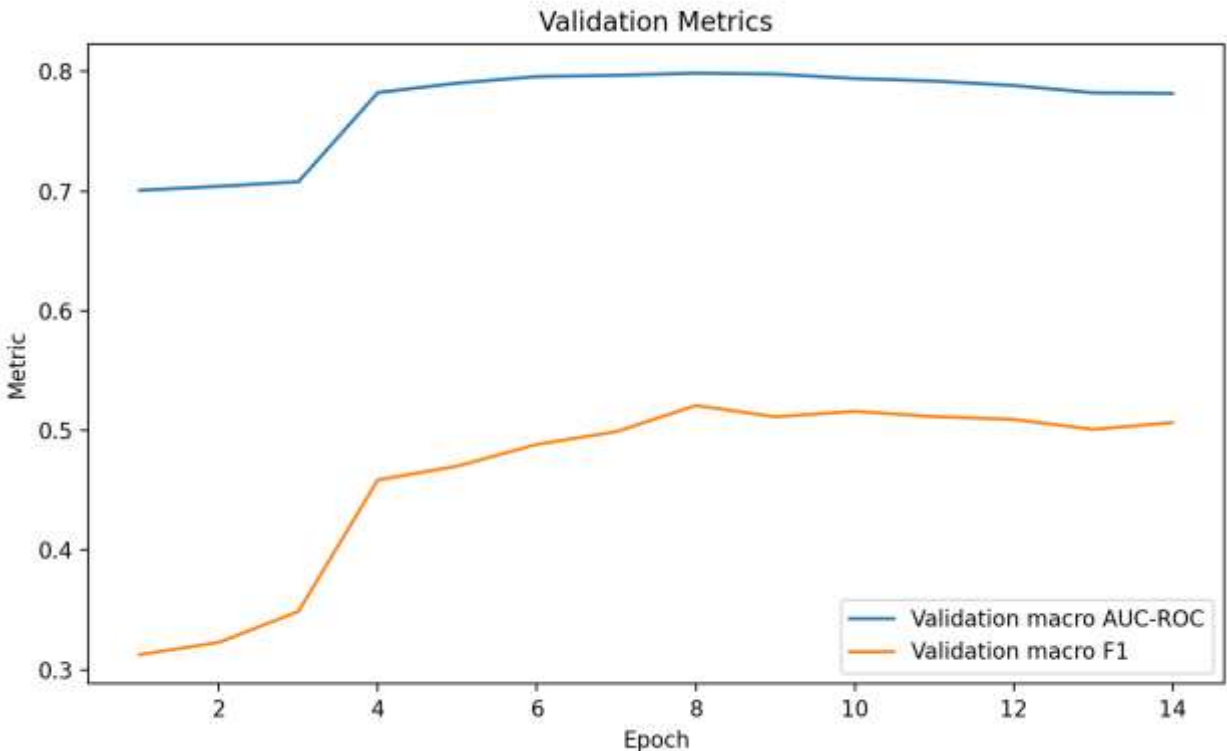
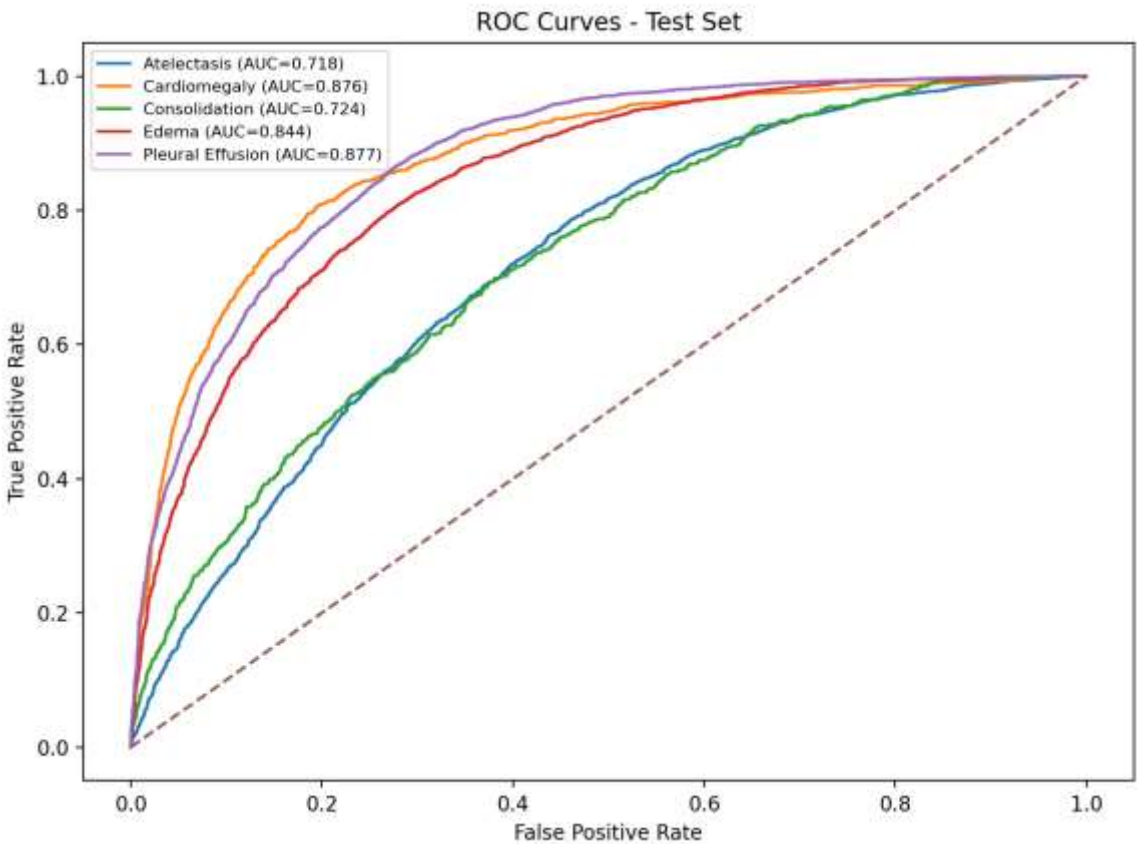
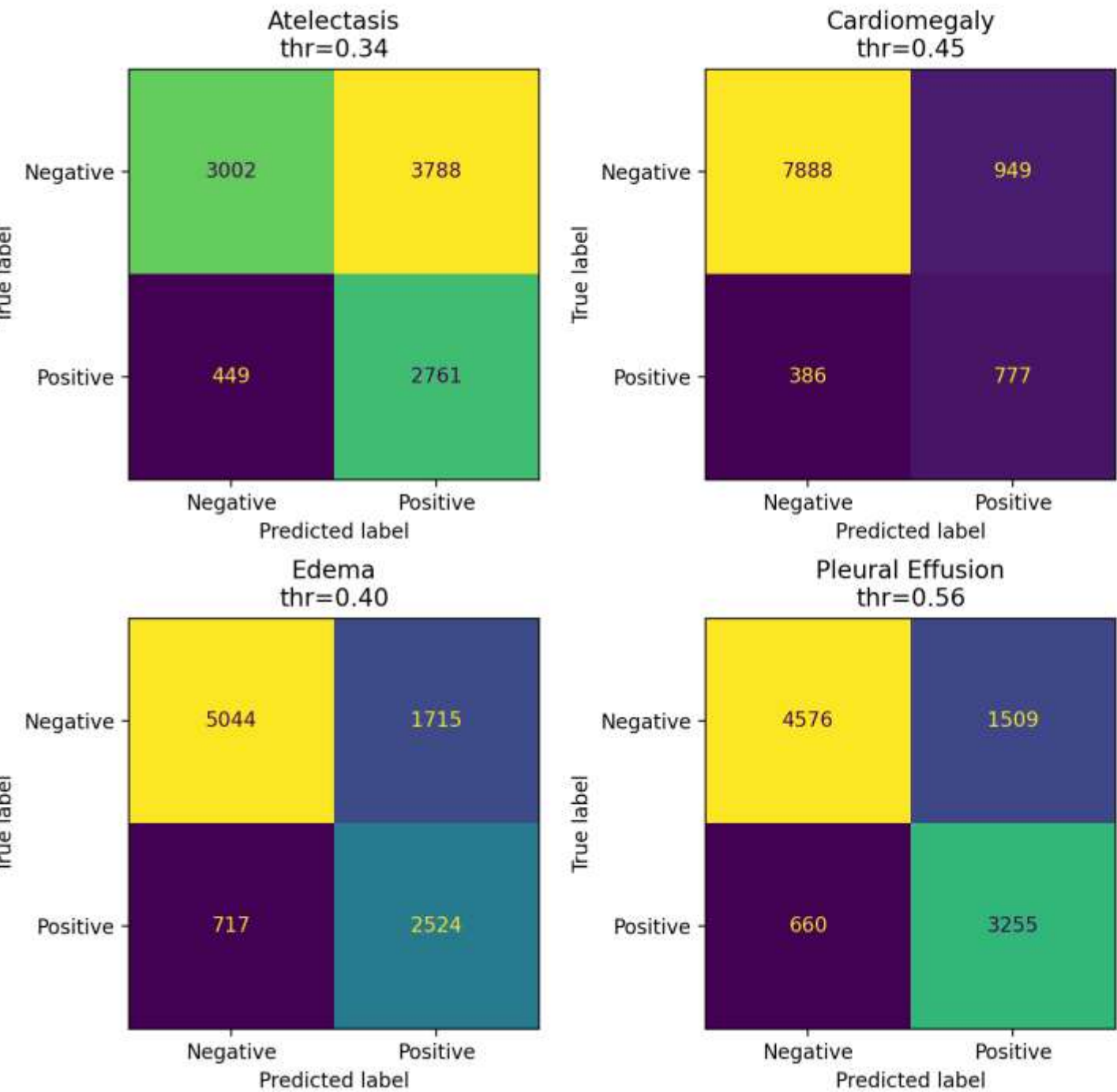
FINAL RESULTS – VARIANTS ENSEMBLE

Evaluation Mode	Macro AUC-ROC	Macro AUC-PR	Macro F1	Recall
Threshold 0.5	0.810107	0.551118	0.534912	0.582244
Optimized Thresholds	0.810107	0.551118	0.554768	0.709403

i Optimized thresholds significantly boost **Recall (+12.7%)** and **F1-score**, facilitating better triage performance.

CONFUSION MATRIX & ROC ANALYSIS

Confusion Matrices by Class - Test Set

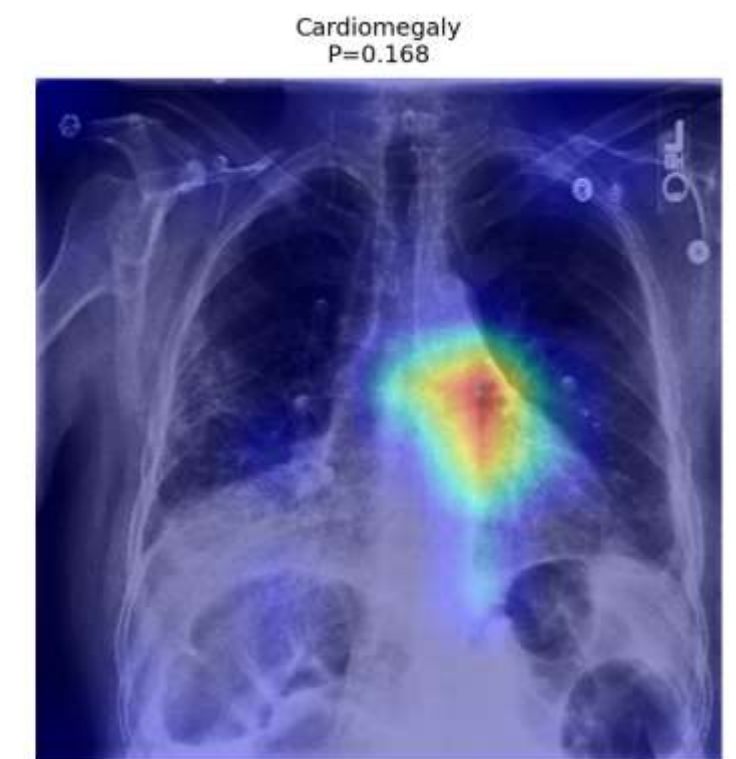


Consolidation remains the most challenging class due to low label frequency and visual subtlety.

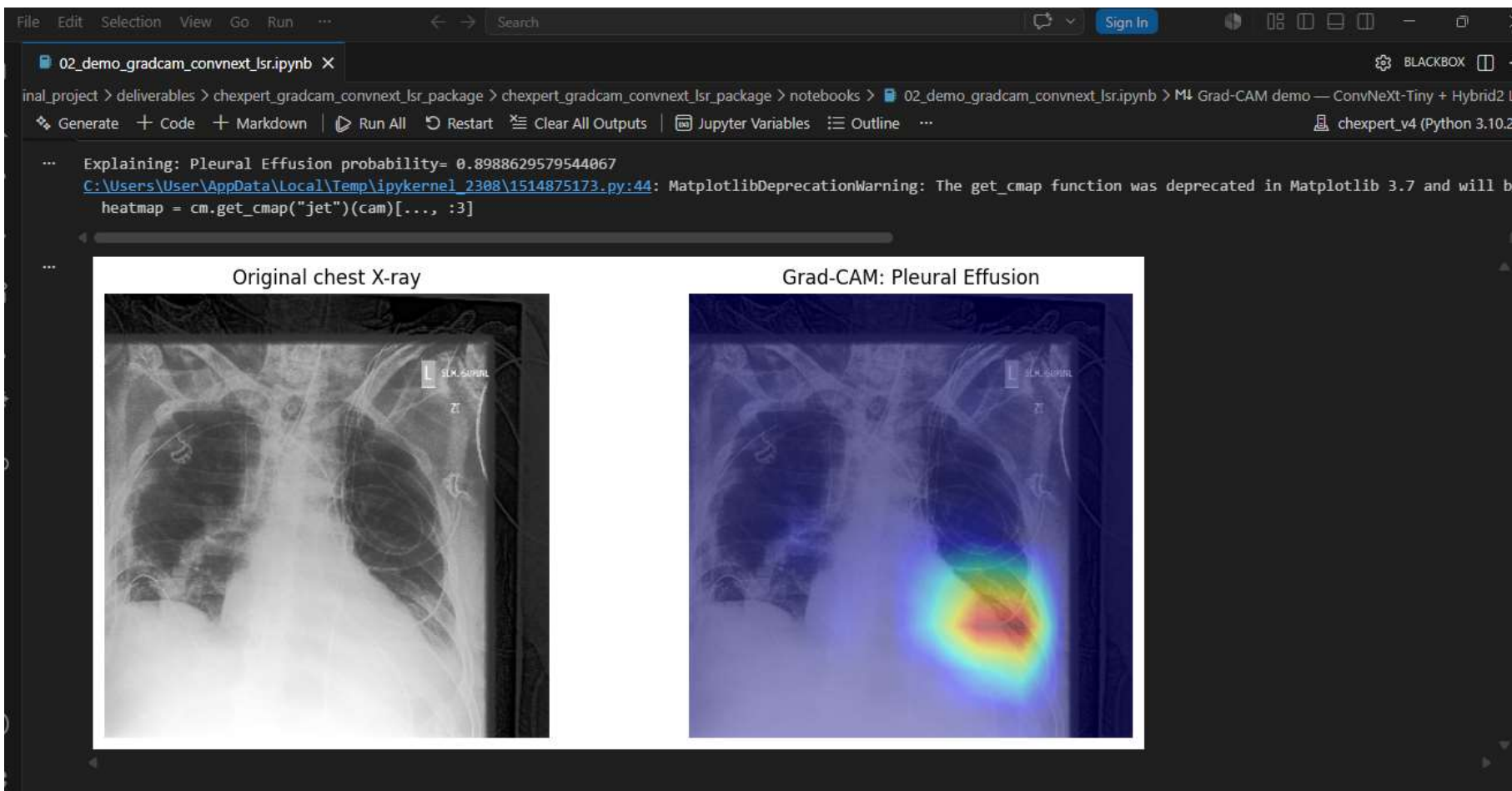
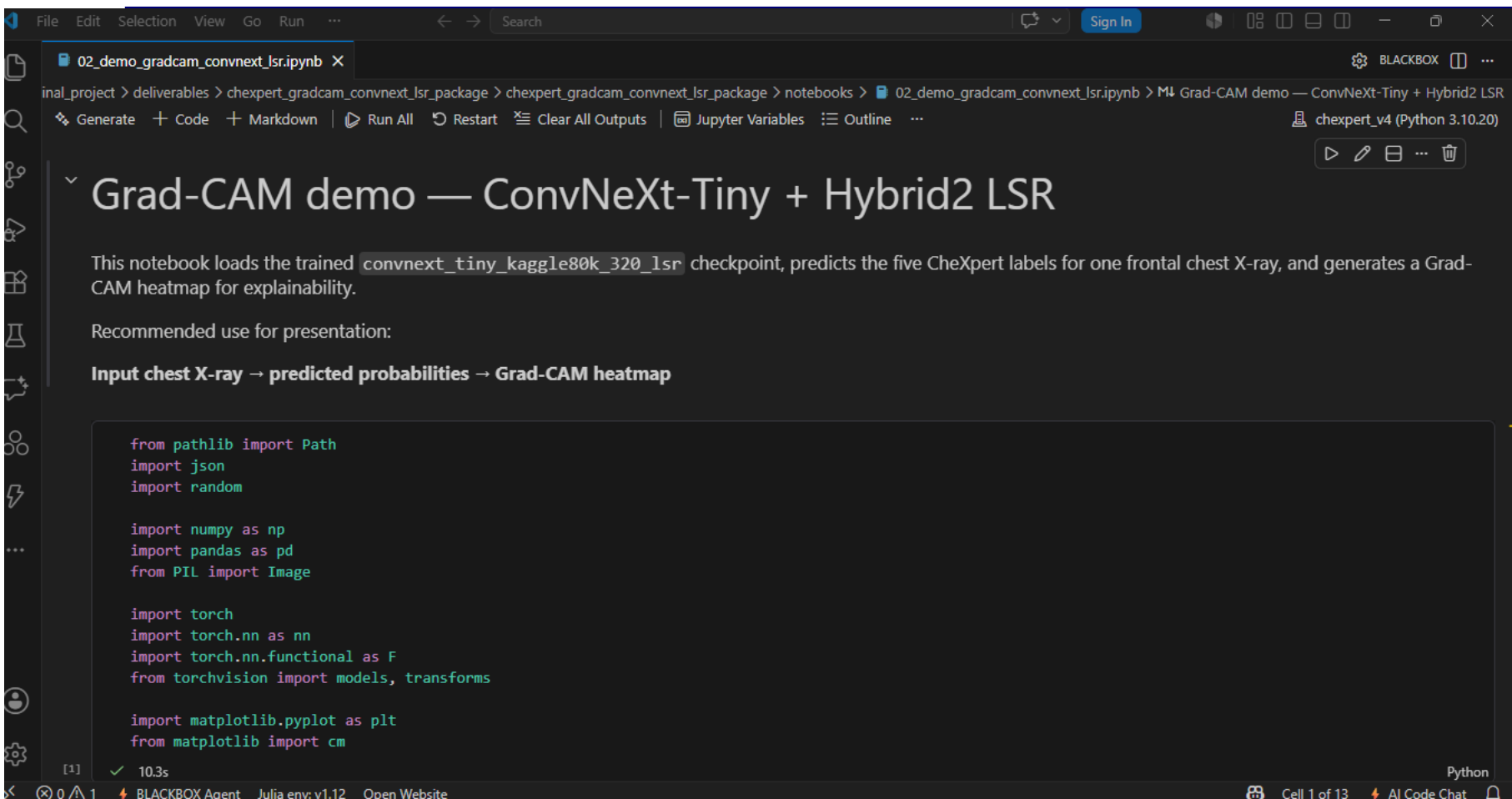
GRAD-CAM EXPLAINABILITY

- Visual localization of prediction drivers.
- Applied to the LSR variant for visualization.
- Confirms model focus on clinically relevant anatomy (e.g., costophrenic angles for Pleural Effusion).
- Used for clinical sanity-checking of predictions.

Case Study: True Positive Findings



LIVE INFERENCE DEMONSTRATION



INTERACTIVE PIPELINE

- Selection of validation CXR sample.
- Real-time preprocessing & prediction.
- Probability heatmap generation.
- Comparison with Ground Truth labels.

	label	probability	threshold	prediction	true_label
4	Pleural Effusion	0.898863	0.56	True	1
0	Atelectasis	0.626019	0.34	True	0
2	Consolidation	0.330785	0.35	False	0
3	Edema	0.126118	0.40	False	0
1	Cardiomegaly	0.123743	0.45	False	0

PROJECT REPOSITORY

Drive link 

<https://drive.google.com/drive/folders/1xFy8NUJ0nKHwxgmHZoeJb75N-8sC99OV?usp=sharing>

CONCLUSIONS

- Architecture Performance: The ConvNeXt-Tiny ensemble provides a strong Macro AUC-ROC of 0.810.
- Triage Optimization: Threshold tuning is essential for clinical applicability, achieving 70.9% Recall.
- Reliability: Patient-wise splitting ensures the model learns general features rather than patient-specific traits.
- Transparency: Grad-CAM integration validates that the model attends to standard radiological regions of interest.

REFERENCES

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